

# Convex Optimization

## Lecture 5: Convex Programming and Convex Set

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# Outline

- 1 Opening Example
- 2 Convex Set
- 3 Operations that Maintain Set's Convexity
- 4 Extending Function Maintaining its Convexity

# QP with Equation Constraints Only (1)

- There is an 8 meters long string, we want to cut it into three segments. Each segment will be used to enclose a circle. So how to cut them that minimizes the sum of three circle areas.
- ❶ Given the lengths of three segments are  $x_1$ ,  $x_2$ , and  $x_3$
- ❷  $x_1 + x_2 + x_3 = 8$
- ❸  $r_1 = \frac{x_1}{2\pi}$
- ❹ We want to minimize  $\frac{x_1^2}{4\pi} + \frac{x_2^2}{4\pi} + \frac{x_3^2}{4\pi}$
- ❺ Namely, we minimize  $x_1^2 + x_2^2 + x_3^2$  under the constraint  $x_1 + x_2 + x_3 = 8$

## QP with Equation Constraints Only (2)

① Minimize  $x_1^2 + x_2^2 + x_3^2$  under the constraint  $x_1 + x_2 + x_3 = 8$

$$\begin{array}{ll} \text{Max.} & x_1^2 + x_2^2 + x_3^2 \\ \text{s.t.} & x_1 + x_2 + x_3 = 8 \end{array} \quad (1)$$

# How to solve

- Given  $b = 8, x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, H = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}, c = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, A = [1 \ 1 \ 1]$
- We have

$$\begin{aligned} \text{Min. } & \frac{1}{2} x^T H x + c^T x \\ \text{sub. to } & Ax = b \end{aligned} \quad (2)$$

- How to solve this problem??

# Solve QP by Lagrangian Multiplier (1)

$$\begin{aligned} \text{Min. } & \frac{1}{2}x^T Hx + c^T x \\ \text{sub. to } & Ax = b \end{aligned} \tag{4}$$

- Define Lagrangian function

$$L(x, \lambda) = x^T Hx + c^T x + \lambda(Ax - b) \tag{5}$$

# Solve QP by Lagrangian Multiplier (2)

- Given Lagrangian function

$$L(x, \lambda) = x^T H x + c^T x + \lambda(Ax - b) \quad (5)$$

- Take partial derivative on  $x$  and  $\lambda$

$$\nabla L_x = 0, \nabla L_\lambda = 0 \quad (6)$$

$\Downarrow$

$$\begin{cases} Hx + c + A\lambda = 0 \\ Ax - b = 0 \end{cases} \quad (7)$$

## Solve QP by Lagrangian Multiplier (3)

$$\begin{cases} Hx + c + A\lambda = 0 \\ Ax - b = 0 \end{cases} \quad (7)$$

$$\Downarrow$$

$$\begin{bmatrix} H & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} -c \\ b \end{bmatrix} \quad (8)$$

$$\Downarrow$$

$$\begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} H_{3 \times 3} & A_{3 \times 1}^T \\ A_{1 \times 3} & 0 \end{bmatrix}^{-1} \begin{bmatrix} -c_{3 \times 1} \\ b_{1 \times 1} \end{bmatrix} \quad (9)$$



## Solve QP by Lagrangian Multiplier (4)

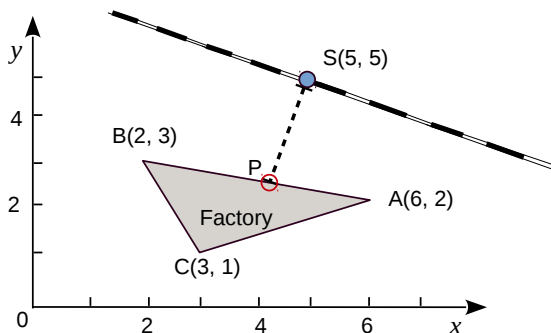
$$\begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} H_{3 \times 3} & A_{3 \times 1}^T \\ A_{1 \times 3} & 0 \end{bmatrix}^{-1} \begin{bmatrix} -c_{3 \times 1} \\ b_{1 \times 1} \end{bmatrix} \quad (10)$$

$$\Downarrow$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \lambda \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & 2 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2.667 \\ 2.667 \\ 2.667 \\ -5.333 \end{bmatrix}$$

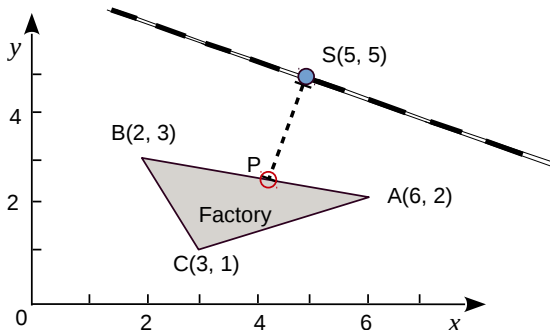
# QP with Equation Constraints and Inequation Constraints

- A factory is going to build a road to train station, the locations of the station and the factory are given in Fig. 1. Please help to find the position  $\mathbf{P}$  within the factory area that connects to the station. Such that the distance between  $\mathbf{P}$  and the train station is the shortest.



# Modeling (1)

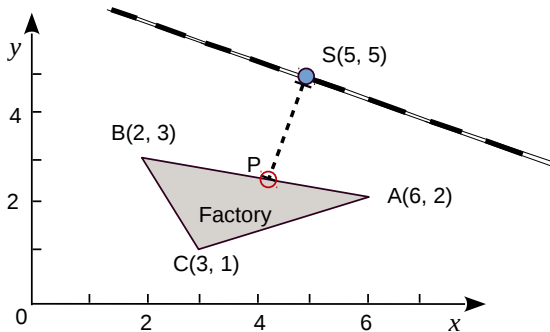
- Let's first look at the target



- Target: Min.  $(x_p - 5)^2 + (y_p - 5)^2$

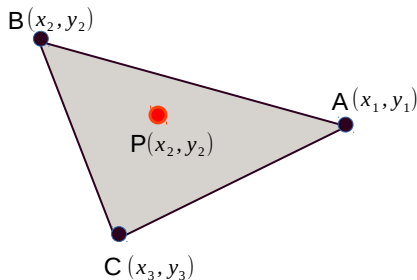
# Modeling (2)

- So what is  $\mathbf{P}$



- $\mathbf{P}$  should be within the factory area

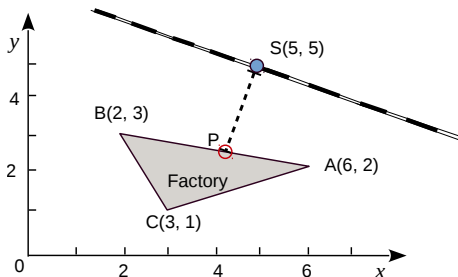
# Convex Set (1)



- Area formed by  $A$ ,  $B$  and  $C$  is convex
- We therefore have followig property, any point  $P(x_p, y_p)$ 
  - ①  $P = \alpha_1 \cdot A + \alpha_2 \cdot B + \alpha_3 \cdot C$ ,  
where  $\alpha_1 + \alpha_2 + \alpha_3 = 1$

# Modeling (3)

- **P** could be expressed as a linear combination of A, B and C



- Namely,  $P = \alpha_1 \cdot A + \alpha_2 \cdot B + \alpha_3 \cdot C = \alpha_1 \begin{bmatrix} 6 \\ 2 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \alpha_3 \begin{bmatrix} 3 \\ 1 \end{bmatrix}$
- $\alpha_1 + \alpha_2 + \alpha_3 = 1$

## Modeling (4)

$$\begin{bmatrix} x_p \\ y_p \end{bmatrix} = \alpha_1 \begin{bmatrix} 6 \\ 2 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \alpha_3 \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

- Plug the expressions for  $x_p$  and  $y_p$  into the target

$$\begin{aligned} \min. \quad & (6\alpha_1 + 2\alpha_2 + 3\alpha_3 - 5)^2 + (2\alpha_1 + 3\alpha_2 + \alpha_3 - 5)^2 \\ \text{sub. to } & \begin{cases} \alpha_1 + \alpha_2 + \alpha_3 = 1 \\ \alpha_1, \alpha_2, \alpha_3 \geq 0 \end{cases} \end{aligned} \quad (2)$$

## Modeling (5)

$$\begin{aligned} \min. \quad & (6\alpha_1 + 2\alpha_2 + 3\alpha_3 - 5)^2 + (2\alpha_1 + 3\alpha_2 + \alpha_3 - 5)^2 \\ \text{sub. to } & \begin{cases} \alpha_1 + \alpha_2 + \alpha_3 = 1 \\ \alpha_1, \alpha_2, \alpha_3 \geq 0 \end{cases} \end{aligned} \quad (2)$$

- Re-organize above model, we have

$$\begin{aligned} \min. \quad & 40\alpha_1^2 + 13\alpha_2^2 + 10\alpha_3^2 + 36\alpha_1\alpha_2 \\ & + 40\alpha_1\alpha_3 + 18\alpha_2\alpha_3 - 80\alpha_1 - 50\alpha_2 - 40\alpha_3 \\ \text{sub. to } & \begin{cases} \alpha_1 + \alpha_2 + \alpha_3 = 1 \\ \alpha_1, \alpha_2, \alpha_3 \geq 0 \end{cases} \end{aligned}$$



# Modeling (6)

- To further simplify the model

$$\begin{aligned} \min. \quad & 40\alpha_1^2 + 13\alpha_2^2 + 10\alpha_3^2 + 36\alpha_1\alpha_2 \\ & + 40\alpha_1\alpha_3 + 18\alpha_2\alpha_3 - 80\alpha_1 - 50\alpha_2 - 40\alpha_3 \\ \text{sub. to } & \begin{cases} \alpha_1 + \alpha_2 + \alpha_3 = 1 \\ \alpha_1, \alpha_2, \alpha_3 \geq 0 \end{cases} \end{aligned}$$

$\Downarrow$

$$\begin{aligned} \min. \quad & [\alpha_1 \ \alpha_2 \ \alpha_3] \begin{bmatrix} 40 & 18 & 20 \\ 18 & 13 & 9 \\ 20 & 9 & 10 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} + \begin{bmatrix} -80 \\ -50 \\ -40 \end{bmatrix}^T \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} \\ \text{sub. to } & \begin{cases} \alpha_1 + \alpha_2 + \alpha_3 = 1 \\ \alpha \succeq 0 \end{cases} \end{aligned} \quad (3)$$

# The model

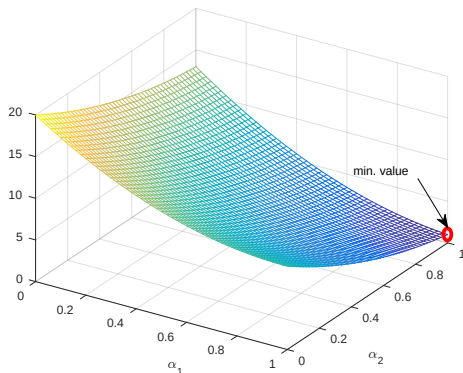
$$\begin{aligned} \min. \quad & [\alpha_1 \ \alpha_2 \ \alpha_3] \begin{bmatrix} 40 & 18 & 20 \\ 18 & 13 & 9 \\ 20 & 9 & 10 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} + \begin{bmatrix} -80 \\ -50 \\ -40 \end{bmatrix}^T \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} \\ \text{sub. to } & \begin{cases} \alpha_1 + \alpha_2 + \alpha_3 = 1 \\ \alpha \succeq 0 \end{cases} \end{aligned} \quad (3)$$

- Two observations
  - 1 The target function is quadratic
  - 2 The constraint is a linear equation<sup>1</sup>.
- This is a typical **quadratic programming** problem

<sup>1</sup>In practice, there could be multiple equations

# Visualize the model

$$\begin{aligned} \min. \quad & [\alpha_1 \ \alpha_2 \ \alpha_3] \begin{bmatrix} 40 & 18 & 20 \\ 18 & 13 & 9 \\ 20 & 9 & 10 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} + \begin{bmatrix} -80 \\ -50 \\ -40 \end{bmatrix}^T \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} \\ \text{sub. to } & \begin{cases} \alpha_1 + \alpha_2 + \alpha_3 = 1 \\ \alpha \succeq 0 \end{cases} \end{aligned} \quad (3)$$



# How to solve

- Given  $b = 1, \alpha = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}, H = \begin{bmatrix} 40 & 18 & 20 \\ 18 & 13 & 9 \\ 20 & 9 & 10 \end{bmatrix}, c = \begin{bmatrix} -80 \\ -50 \\ -40 \end{bmatrix},$   
 $A = [1 \ 1 \ 1]$
- We have

$$\begin{aligned} \min. \quad & \alpha^T H \alpha + c^T \alpha \\ \text{sub. to} \quad & A \alpha = b \\ & \alpha \succeq 0 \end{aligned} \tag{4}$$

- How to solve this problem??

# Solve QP by Lagrangian Multiplier

$$\begin{aligned}
 \min. \quad & \alpha^T H \alpha + c^T \alpha \\
 \text{sub. to} \quad & A \alpha = b \\
 & I \alpha \succeq 0
 \end{aligned} \tag{4}$$

- Define Lagrangian function

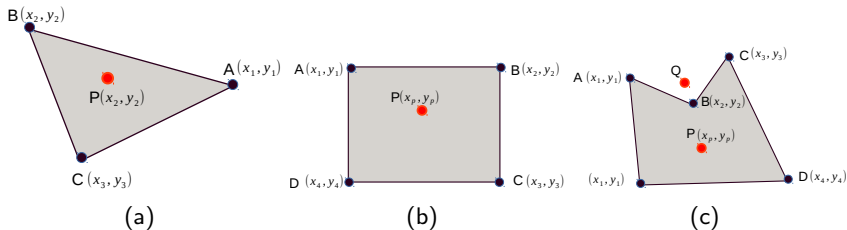
$$L(\alpha, \lambda, \beta) = \alpha^T H \alpha + c^T \alpha + \lambda(A\alpha - b) + \beta^T(I\alpha - 0) \tag{5}$$

- This problem cannot be solved in the analytic way
- Before we start to solve the QP with equations and inequations
- We are going to introduce you systematically the **convex set**, **convex function**, and **convex problem**

# Outline

- 1 Opening Example
- 2 Convex Set
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- 4 Extending Function Maintaining its Convexity

# Convex Set: the definition (1)



- $\forall x_1, x_2 \in C$ , given  $x_i = \theta x_1 + (1 - \theta)x_2$ ,  $\theta \in [0, 1]$ , a set  $C$  is **convex** iff  $x_i \in C$

# Convex Set: the definition (2)

- $\forall x_1, x_2 \in C$ , **given**  $x_i = \theta x_1 + (1 - \theta)x_2$ ,  $\theta \in [0, 1]$ , **a set  $C$  is convex iff**  $x_i \in C$
  - Now, given  $\forall x_1, x_2, \dots, x_k \in C$  and  $C$  is convex,  
 $\theta_1 + \theta_2 + \dots + \theta_k = 1$ ,  $\theta_k \geq 0, k \geq 2$ ,  $x_j = \theta_1 x_1 + \dots + \theta_k x_k$  prove  
 $x_j \in C$
- 1 When  $k = 2$ , it holds according to the definition of convex set
  - 2 When  $k \geq 2$ , we prove the case when  $k = 3$ , and extends it to  $k > 3$

Think about it in 5 minutes...



# Convex Set: the definition (3)

- Given convex set  $C$ , we have  $\forall x_1, x_2, x_3 \in C$ . We also have  $\theta_1 + \theta_2 + \theta_3 = 1, \theta_1, \theta_2, \theta_3 \in [0, 1]$ , we are going to prove  $x_i \in C$  when  $x_i = \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$

- According to the definition of convex set, we have

$$\frac{\theta_1}{\theta_1 + \theta_2} x_1 + \frac{\theta_2}{\theta_1 + \theta_2} x_2 \in C$$

- Therefore, we can build another convex combination

$$x_i = (\theta_1 + \theta_2) \left( \frac{\theta_1}{\theta_1 + \theta_2} x_1 + \frac{\theta_2}{\theta_1 + \theta_2} x_2 \right) + (1 - \theta_1 - \theta_2) x_3$$

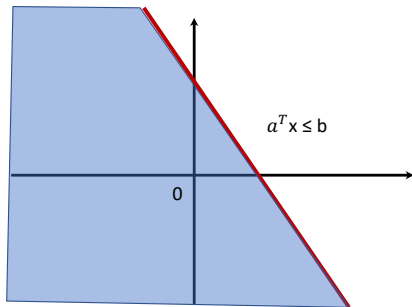
- Since  $x_3 \in C$ ,  $x_i \in C$  as well, to this end, the case is proved when  $k = 3$
- In the same way, we can prove the case when  $k > 3$

# Convex Hull: the definition (1)

- Given a set  $\mathbf{S}$ , the minimal convex set that include set  $S$  is called the convex hull of  $S$ , it is denoted as  $\mathbf{conv}(\mathbf{S})$
- Namely,  $\forall x_1, x_2 \in S, \theta \in [0, 1]$ , we have
$$\theta x_1 + (1 - \theta)x_2 \in \mathit{Conv}(S)$$

# Representative Convex Sets – Half-space

- Given plane  $\{x | a^T x \leq b, x, a \in R^n\}$ , the half-space is a convex set



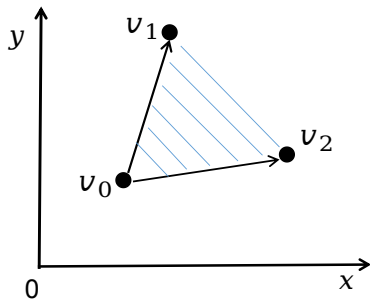
# Representative Convex Sets – Polyhedral

- Given  $a_i, c_j, x \in R^n$ , the polyhedral is defined as

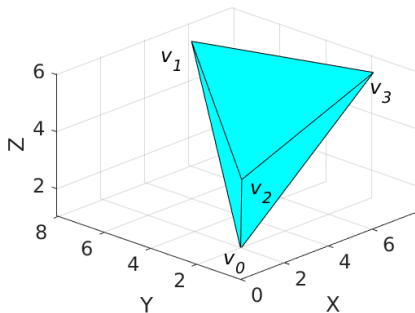
$$P = \left\{ x \mid \begin{array}{l} a_i^T x \leq b_i, i = 1, \dots, m \\ c_j^T x = d_j, j = 1, \dots, p \end{array} \right\}. \quad (6)$$

- One  $a_i^T x \leq b_i$  defines a half-space, which is convex
- One equation  $c_j^T x = d_j$  defines a plane, which is convex as well
- The intersection of convex sets are convex as well

# Representative Convex Sets – Simplex (1)



(d) 2D Simplex



(e) 3D Simplex

- Given  $v_0, v_1, \dots, v_k \in R^n$  and  $k \leq n$ ,  $v_1 - v_0, \dots, v_k - v_0$  are linearly independent
- The simplex is the convex hull of  $v_0, v_1, \dots, v_k$

# Representative Convex Sets – Simplex (2)

- Given  $v_0, v_1, \dots, v_k \in R^n$  and  $k \leq n$ ,  $v_1 - v_0, \dots, v_k - v_0$  are linearly independent
- The simplex is the convex hull (**H**) of  $v_0, v_1, \dots, v_k$
- $\forall x = \theta_0 v_0 + \dots + \theta_i v_i + \dots + \theta_k v_k, \sum_{i=0}^k \theta_i = 1, \theta_i \geq 0$
- $\Rightarrow x \in \mathbf{H}$

In reverse,

- $\forall x \in H$ , it can be expressed as

$$x = \theta_0 v_0 + \dots + \theta_i v_i + \dots + \theta_k v_k, \quad (7)$$

$$\text{where } \sum_{i=0}^k \theta_i = 1, \theta_i \geq 0$$

# Representative Convex Sets – Convex Cone (1)

- Given the set  $x_1, x_2, \dots, x_k \in \mathbf{C}$  and  $\theta_1, \theta_2, \dots, \theta_k \geq 0$
- $\theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k \in \mathbf{C}$
- Please show set  $\mathbf{C}$  is convex

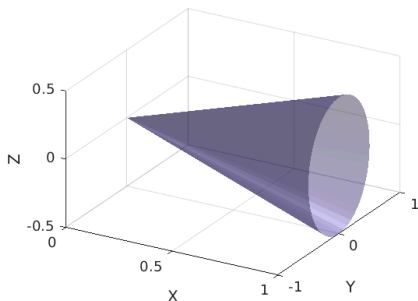
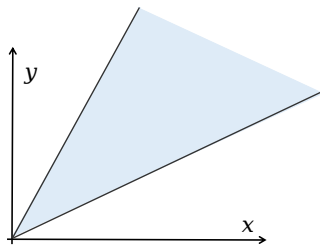


Figure: Convex Cone in 2D and 3D.

Think about it in 5 minutes...

# Representative Convex Sets – Convex Cone (2)

- Given the set  $x_1, x_2, \dots, x_k \in \mathbf{C}$  and  $\theta_1, \theta_2, \dots, \theta_k \geq 0$
- $\theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k \in \mathbf{C}$
- Please show set  $\mathbf{C}$  is convex
- **Proof:**
- Given  $x, y \in \mathbf{C}$ ,  $\lambda \in [0, 1]$ ,  $z = \lambda x + (1 - \lambda)y$
- According to the definition of convex cone,  $z \in \mathbf{C}$  as well
- So  $\mathbf{C}$  is convex

— proved —



# Representative Convex Sets – Symmetric Matrix (1)

- Given the set  $S^n = \{X \in R^{n \times n} | X = X^T\}$ ,  $S$  is convex
- Prove it based on the definition of convex set

Think about it in 5 minutes...

# Representative Convex Sets – Symmetric Matrix (2)

- Given the set  $S^n = \{X \in R^{n \times n} | X = X^T\}$ ,  $S^n$  is convex
- **Proof:**
- Given  $A, B \in S^n, \theta \in [0, 1]$

$$X = \theta A + (1 - \theta)B$$

- Obviously,  $X$  is symmetric and  $X \in R^{n \times n}$

# Representative Convex Sets – Semi-def. Sym. Matrix

- Given the set  $S_+^n = \{X \in R^{n \times n} | X = X^T, X \succeq 0\}$ ,  $S_+^n$  is convex

- Proof:**

- Given  $A, B \in S_+^n, \theta \in [0, 1]$

- $\forall x \in R^n$ , We have  $x^T A x \geq 0, x^T B x \geq 0$

$$x^T A x \geq 0$$

$$x^T B x \geq 0$$

(8)

- So we have

$$\theta x^T A x + (1 - \theta) x^T B x \geq 0$$

$$x^T (\theta A + (1 - \theta) B) x \geq 0$$

(9)

— proved —

# Representative Convex Sets – Definite Sym. Matrix

- Given the set  $S_{++}^n = \{X \in R^{n \times n} | X = X^T, X \succ 0\}$ ,  $S_{++}^n$  is convex
- The proof is quite similar
- We leave it as a homework

# Summary over Convex Sets

| Name               | Expression                                                                                                                      |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Plane              | $a^T x + b = 0$                                                                                                                 |
| Half-space         | $a^T x \leq b$                                                                                                                  |
| Polyhedral         | $P = \left\{ x \mid \begin{array}{l} a_i^T x \leq b_i, i = 1, \dots, m \\ c_j^T x = d_j, j = 1, \dots, p \end{array} \right\}.$ |
| Simplex            | $\text{conv}\{v_0, \dots, v_k\}, \text{rank}([v_1 - v_0; \dots; v_k - v_0]) = k$                                                |
| Symmetric Mat.     | $S^n = \{X \in R^{n \times n} \mid X = X^T\}$                                                                                   |
| S.-D. Symm. Matrix | $S_+^n = \{X \in R^{n \times n} \mid X = X^T, X \succeq 0\}$                                                                    |
| Def. Symm. Matrix  | $S_{++}^n = \{X \in R^{n \times n} \mid X = X^T, X \succ 0\}$                                                                   |

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# Intersection of Two Convex Sets (1)

- Given  $A, B \in R^n$  are convex, then  $A \cap B$  is convex
- How to prove it holds?

Think about it in 5 minutes...

## Intersection of Two Convex Sets (2)

- Given sets  $A, B \in R^n$  are convex, then  $A \cap B$  is convex

- **Proof:**

- Given  $x_1, x_2 \in A \cap B$ ,  $\theta \in [0, 1]$

- $x_i = \theta x_1 + (1 - \theta)x_2$

① Since  $x_1, x_2 \in A$ ,  $A$  is convex, so  $x_i \in A$

② Since  $x_1, x_2 \in B$ ,  $B$  is convex, so  $x_i \in B$

③ As  $x_i$  is shared by  $A$  and  $B$ , it is true that  $x_i \in A \cap B$

— proved —



# Affine Transformation (1)

- Given  $f : R^n \rightarrow R^m$  is an affine transform, namely  $f(x) = Ax + b$
- $A \in R^{m \times n}, b \in R^m$
- If  $S \in R^n$  is convex, then  $Q = \{f(x) | x \in S\}$  is convex as well
- How to prove it??

Think about it in 5 minutes...

## Affine Transformation (2)

- Given  $f : R^n \rightarrow R^m$  is an affine transform, namely  $f(x) = Ax + b$
- $A \in R^{m \times n}, b \in R^m$
- If  $S \in R^n$  is convex, then  $Q = \{f(x) | x \in S\}$  is convex as well
- **Proof:**
- Given  $x, y \in S^n$ , after  $f(\cdot)$  transformation, we have
- $Ax + b$  and  $Ay + b$ , they are inside  $Q$ , we should check whether  $z \in Q$
- $z = \theta(Ax + b) + (1 - \theta)(Ay + b), \theta \in [0, 1]$

$$z = \theta(Ax + b) + (1 - \theta)(Ay + b)$$

$$z = A(\theta x + (1 - y)\theta) + b$$

- Since  $\theta x + (1 - y)\theta \in S$ ,  $z \in Q$  as well

— proved —

# Perspective Transformation (1)

- The perspective transformation is defined as  $P : R^{n+1} \rightarrow R^n$ ,  
 $(\tilde{x}, x_{n+1}), x_{n+1} > 0$ ,
- Namely,  $P(\tilde{x}, x_{n+1}) = \frac{\tilde{x}}{x_{n+1}}, \tilde{x} \in R^n$
- Now let's prove this transformation maintains the convexity of a convex set

Think about it in 5 minutes...

## Perspective Transformation (2)

- Given  $x = (\tilde{x}, x_{n+1})$ , and  $y = (\tilde{y}, y_{n+1})$
- A segment connecting  $x$  and  $y$  is  $\forall \theta \in [0, 1]$
- We should prove that  $\theta x + (1 - \theta)y$  is a segment after the projection

Think about it in 5 minutes...

# Perspective Transformation (3)

- Given  $x = (\tilde{x}, x_{n+1})$ , and  $y = (\tilde{y}, y_{n+1})$
- A segment connecting  $x$  and  $y$  is  $\forall \theta \in [0, 1]$
- We should prove that  $\theta x + (1 - \theta)y$  is a segment after the projection
- **Proof:**

$$\begin{aligned}
 \theta x + (1 - \theta)y &\rightarrow P(\theta x + (1 - \theta)y) \\
 P(\theta x + (1 - \theta)y) &= \frac{\theta \tilde{x} + (1 - \theta)\tilde{y}}{\theta x_{n+1} + (1 - \theta)y_{n+1}} \\
 &= \frac{\theta x_{n+1}}{\theta x_{n+1} + (1 - \theta)y_{n+1}} \cdot \frac{\tilde{x}}{x_{n+1}} + \frac{(1 - \theta)y_{n+1}}{\theta x_{n+1} + (1 - \theta)y_{n+1}} \cdot \frac{\tilde{y}}{y_{n+1}}
 \end{aligned}$$

# Perspective Transformation (4)

- We should prove that  $\theta x + (1 - \theta)y$  is a segment after the projection

- **Proof:**

$$\theta x + (1 - \theta)y \rightarrow P(\theta x + (1 - \theta)y)$$

$$\begin{aligned} P(\theta x + (1 - \theta)y) &= \frac{\theta \tilde{x} + (1 - \theta)\tilde{y}}{\theta x_{n+1} + (1 - \theta)y_{n+1}} \\ &= \frac{\theta x_{n+1}}{\theta x_{n+1} + (1 - \theta)y_{n+1}} \cdot \frac{\tilde{x}}{x_{n+1}} + \frac{(1 - \theta)y_{n+1}}{\theta x_{n+1} + (1 - \theta)y_{n+1}} \cdot \frac{\tilde{y}}{y_{n+1}} \end{aligned}$$

- Let  $\mu = \frac{\theta x_{n+1}}{\theta x_{n+1} + (1 - \theta)y_{n+1}}$ . Note that  $0 \leq \mu \leq 1$ , we have

$$= \mu \cdot \frac{\tilde{x}}{x_{n+1}} + (1 - \mu) \cdot \frac{\tilde{y}}{y_{n+1}}$$

$$P(\theta x + (1 - \theta)y) = \mu \cdot P(x) + (1 - \mu) \cdot P(y)$$

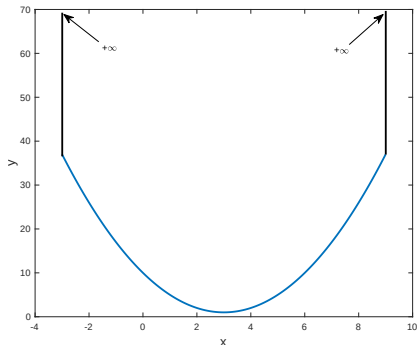
# Outline

- 1 Opening Example
- 2 Convex Set
- 3 Operations that Maintain Set's Convexity
- 4 Extending Function Maintaining its Convexity**

# Function Extension Maintaining its Convexity (1)

- Given function  $f : R^n \rightarrow R$ , it is a convex function,  $domf = C \subseteq R^n$
- Now, we extend it to the whole  $R^n$  and maintaining its convexity

$$\widetilde{f(x)} = \begin{cases} f(x) & x \in domf \\ +\infty & x \notin domf \end{cases} \quad (10)$$

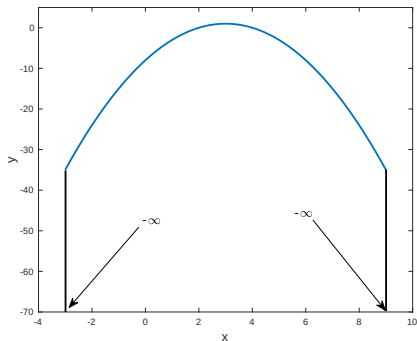




# Function Extension Maintaining its Convexity (2)

- Given function  $f : R^n \rightarrow R$ , it is a concave function,  $domf = C \subseteq R^n$
- Now, we extend it to the whole  $R^n$  and maintaining its convexity

$$\widetilde{f}(x) = \begin{cases} f(x) & x \in domf \\ -\infty & x \notin domf \end{cases} \quad (11)$$



# Indicator Function

- Given convex set  $C \in R^n$ , and the indicator function defined on it

$$f_C = \begin{cases} \text{undefined} & x \notin C \\ 0 & x \in C \end{cases} \quad (12)$$

- It can be extended to the whole  $R^n$  while maintaining its convexity

$$f_C = \begin{cases} +\infty & x \notin C \\ 0 & x \in C \end{cases} \quad (13)$$